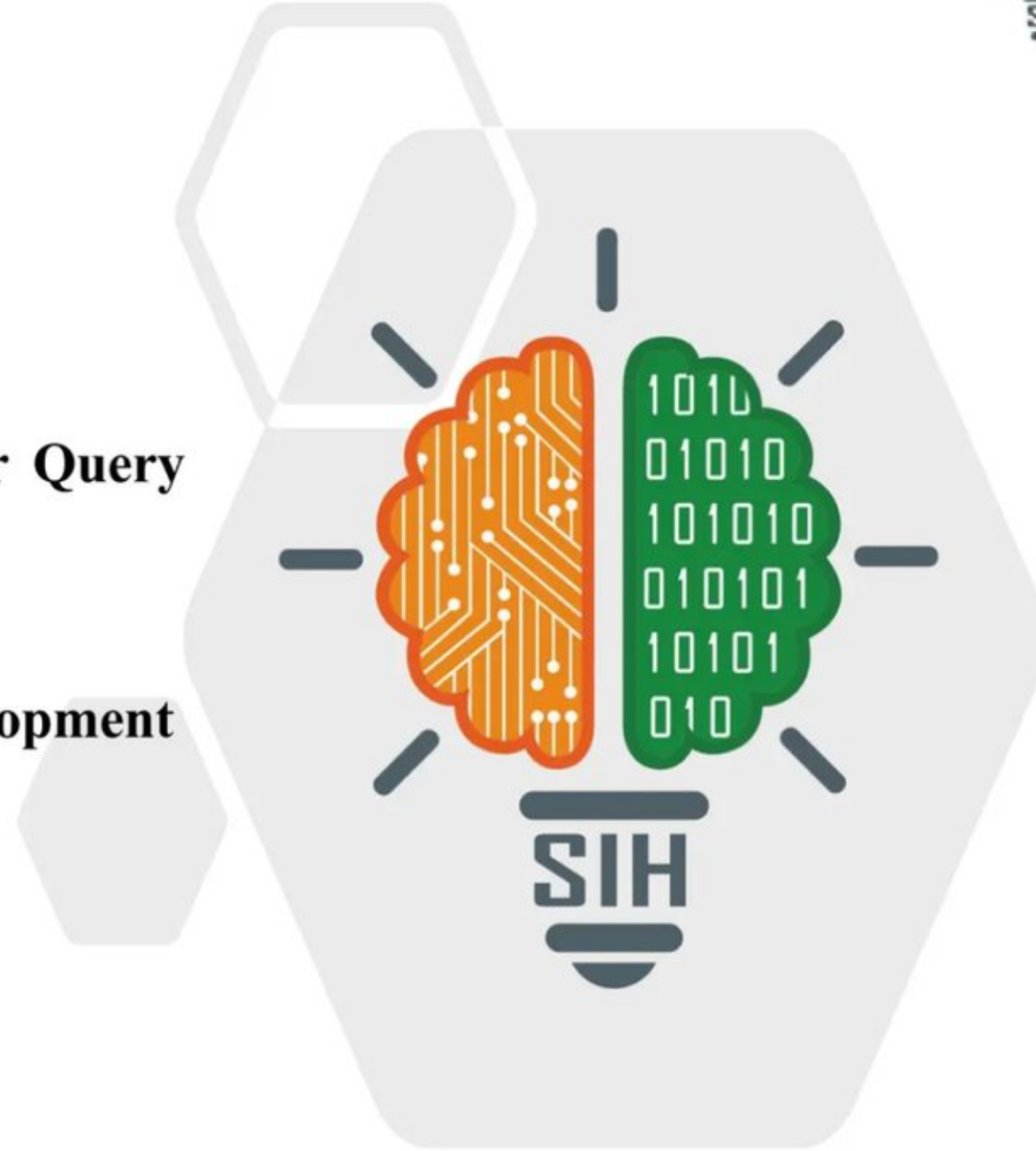


SMART INDIA HACKATHON 2025



- **Problem Statement ID –SIH25076**
- **Problem Statement Title- AI-Based Farmer Query Support and Advisory System**
- **Theme-Agriculture, FoodTech & Rural Development**
- **PS Category- Software**
- **Team ID-59272**
- **Team Name- IT CRACKERS**



IDEA/APPROACH DETAILS

PROPOSED SOLUTION

We aim to develop an **AI-powered Digital Krishi Officer** app to provide farmers with **instant, personalized, and reliable agricultural support** in **Malayalam and other regional languages**.

KEY FEATURES

- **Multilingual Inputs** – Support **voice, text & image queries** in **local languages**.
- **IoT Sensors** – Capture **real-time farm data** (soil, weather, crops).
- **AI Disease Detection** – Identify **crop diseases** early with machine learning.
- **Smart Queries** – Give **personalized answers** using farm context.
- **Expert Escalation** – Route complex issues to **agriculture experts**.
- **Offline Support** – Works in **low network areas** with cached data.
- **Subsidy & Market Info** – Updates on **schemes** and **market prices**.
- **Dashboards & Alerts** – Provide **farm insights** through **notifications**.
- **Secure Storage** – Follow **OWASP** standards for **data protection**.
- **Trend Analysis** – Show **farm health patterns** and predictions.



Why This Stands Out?

☐ AI + IoT Intelligence

☒ AI + Human Hybrid Escalation

☐ Works in Low-Connectivity Areas

TECHNOLOGIES USED:

ResNet-50 : Image analysis for plant disease detection

AssemblyAI API: Transcribe farmer audio inputs

NLP (Natural Language Processing): Analyze transcribed audio for symptom detection

OpenWeatherMap API: Retrieve GPS-based weather data

IoT Sensors (Soil moisture, DHT11, LDR, pH): Monitor land and crop conditions

Microsoft Translator API: Multilingual translation of diagnoses and solutions

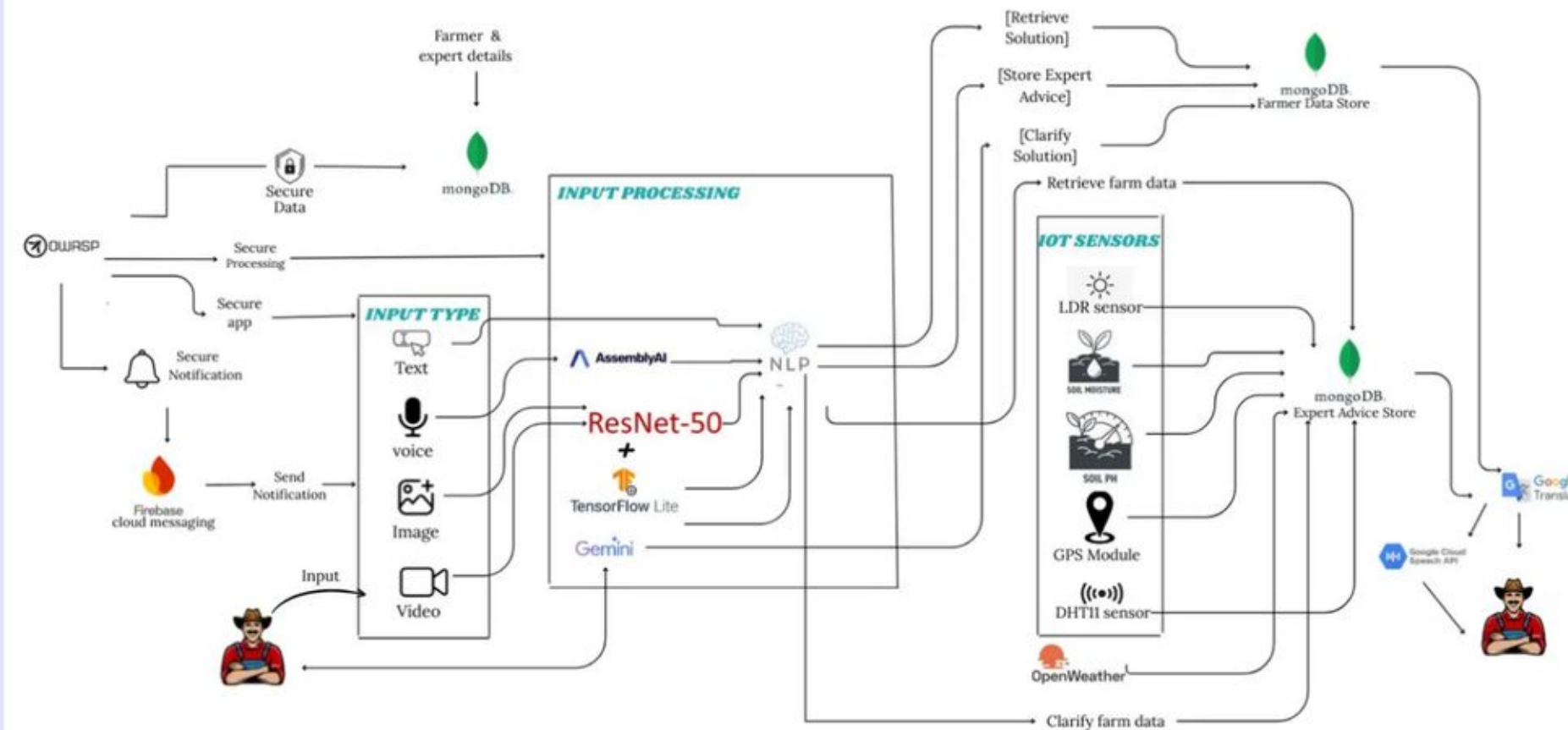
TensorFlow Lite: Offline disease detection on mobile devices

Firebase Cloud Messaging: Push notifications for reminders and alerts

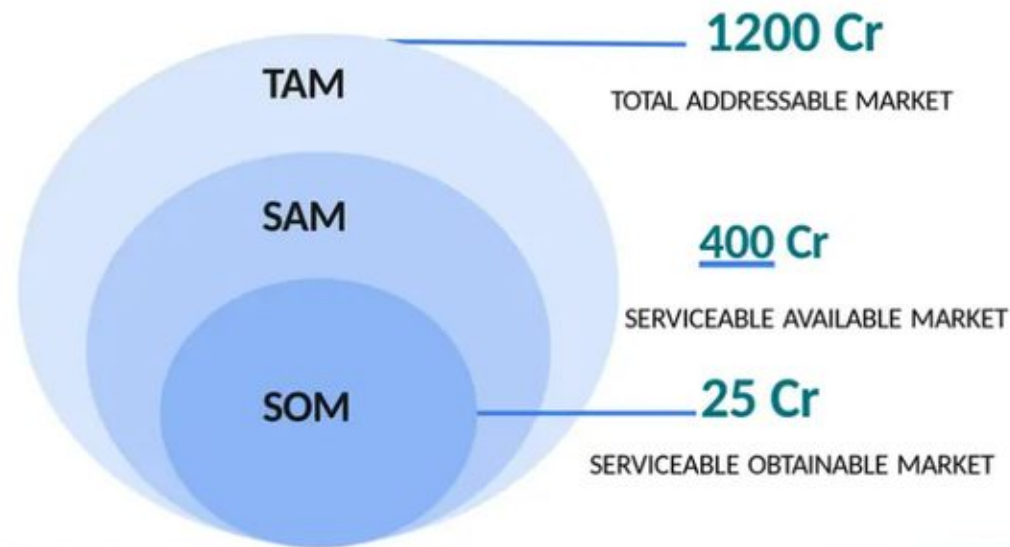
Agrobot (Automation Hardware): Execute automated farming tasks

GPS Module: Track location for weather and farming data

OWASP Security: Ensure app security and vulnerability protection



Market Size



Calculations

Cost Structure & Revenue Streams

Technology & Development Costs: Mobile app, AI/ML training (ResNet-50, NLP, TensorFlow Lite), IoT sensor integration, cloud hosting.

Operations & Support Costs: Agri experts, data annotation, multilingual integration, expert escalation support.

Partnerships: With **agri universities**, **govt. schemes**, **NGOs** for scaling adoption.

Revenue Streams:

Govt. subsidies for digital advisory adoption.
CSR funding for large-scale farmer support.

Challenges

Solutions

Data Scarcity in Local Crops

There is a **lack of quality datasets** for regional crops and pests, so we will **partner with agri universities** and use **farmer-uploaded images** to continuously improve our models.

Scalability of AI Models

Challenge: Farmers speak diverse local languages, many in low-network areas.
Solution: Microsoft Translator API + NLP + AssemblyAI for multilingual support. Enable offline mode with cached data for low-connectivity regions.

Monetization & Affordability

• **Challenge:** Farmers are price-sensitive, limiting adoption.
• **Solution:** Introduce **freemium model** with free basic advisory. Advanced insights funded by **govt. subsidies**, **agri co-ops**, **NGOs**, and **CSR initiatives**.

- **Bias Mitigation in AI Tools**

The platform eliminates bias by delivering **data-backed, region-specific** agricultural advice in **local languages**, ensuring that all farmers—regardless of location or literacy—receive equal support.

- **Real-Time & Personalize Guidance**

AI offers **personalized guidance** tailored to each farmer's specific **crop type, soil condition, location, and season**, ensuring that the advice is not only relevant but also **actionable and timely**.

- **Inclusive Rural Access**

By supporting **voice, text, and image inputs** in **multiple regional languages**, the solution empowers **low-literacy** and **non-English-speaking** farmers, bridging the digital divide in rural communities.

- **Farmer Empowerment**

The app is designed to be **affordable, scalable**, and usable in **low-connectivity environments**, making expert agricultural advice accessible to **farmers who need it the most**.

- **Eco-Friendly & Sustainable Practice**

Eco-friendly farming practices are promoted through **AI + IoT** intelligence, supporting climate-resilient and water-efficient agriculture. This ensures long-term sustainability while reducing input costs for farmers.

- **Continuous Learning & Adaptation**

The AI engine improves continuously through **farmer feedback** and **expert input**, making future responses more accurate, localized, and effective over time.

Industry Outlook

78%

of farmers in India rely on traditional methods of getting advice (neighbors, local shops).

65%

smartphone penetration in rural areas (2025).

72%

farmers face crop losses due to lack of timely expert advice.

15%

annual rise in agri-tech app adoption.

Primary Research

Research Sample:
Interviews (Farmers: 30, Agri
Officers: 10), Google Form
Responses: 120

74%

farmers want **regional language support** for advice.

69%

feel **AI-based crop & pest detection** will save time and cost.

83%

prefer **voice or image-based query support**.

References

- FAO (2024): Digital Agriculture Report
→ [FAO Digital Agriculture](#)
- NITI Aayog (2023): Agri-Tech in India Study
→ [NITI Aayog Reports](#)
- World Bank (2023): AI in Agriculture Insights
→ [World Bank Agriculture & AI](#)
- AgFunder (2024): India Agri-Tech Market Report
→ [AgFunder Reports](#)

Farmers Insights

Difficulty accessing trusted crop advisory.

No timely alerts on pest/disease outbreaks.

Limited awareness of government schemes.

Need guidance on eco-friendly practices

Agri Officer Insights

High workload handling repetitive farmer queries.

Lack of AI tools to filter & prioritize cases.

Limited manpower to cover all villages

Need smart escalation system for complex cases.

Market Voice

"I struggle to get advice in my language."-
- Farmer, Kerala

"Farmers don't follow scientific practices due to lack of real-time support."
- Agri Officer

"AI apps can reduce crop loss, but they must be easy to use."
- Agri Expert

"Most apps are hard for low-literate farmers."
- Farmer Leader